

An Endogenous Moral Formation Game^{*}

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Abstract

This paper develops a model of endogenous moral formation. In a two-stage game, players first choose *moral functions* assigning *value judgments* to actions, then select actions. Two motives drive moral choice: the *influencing-outcome motive* leads players to praise actions that benefit themselves; the *self-righteous motive* leads them to praise their own planned actions. Large moral weights support any homogeneous outcome through self-fulfilling moral functions; heterogeneous outcomes require that no homogeneous alternative strictly Pareto dominates them. Applications include cooperation in the Prisoners Dilemma, equilibrium selection in coordination games, status-seeking consumption, and internalization of values that favor powerful players.

Keywords: social norms, coordination games, equilibrium selection, externalities, cooperation

JEL Codes: C72, D64, D62, Z13

How are moral standards formed, and why do they take such different shapes across societies, historical periods, and social groups? And why do individuals sometimes adopt moral standards that appear to serve others' interests more than their own? This paper develops a model of endogenous moral formation to address these questions. A *moral game* is a two-stage game built on top of a standard material game. In stage 1, players simultaneously choose *moral functions* that assign *value judgments* to actions. In stage 2, players observe all chosen moralities and select actions. Value judgments enter payoffs directly through *moral weights* w_{ij} , which measure how much player i cares about player j 's evaluation of i 's action.

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Because moral functions are chosen before actions are taken, they are credible commitments rather than ex-post rationalizations.

My modelling innovation is to grant agents full freedom over their moral standards, including what they praise and what they condemn, and to use that freedom to rigorously link social ethics with underlying material interests. Existing models of social preferences treat the direction and intensity of moral concern as exogenous parameters, fixed features of preferences that constrain behavior. This paper inverts that approach. Agents endogenously choose what to praise and condemn, and those choices are disciplined by material payoffs. The content of ethical norms is derived from, and held accountable to, the material stakes of the agents forming them. This inversion transforms morality from a constraint on behavior into a strategic instrument for shaping it, and allows the model to generate moral diversity, norm formation, and value internalization as equilibrium phenomena, without imposing any prior moral content.

Two motives drive the choice of moral functions. The *influencing-outcome motive* leads individuals to praise actions that benefit themselves, creating incentives for others to take those actions. The *self-righteous motive* leads individuals to assign high value judgments to the actions they plan to take, gaining social payoff from self-approval. Both motives operate simultaneously and interact to determine equilibrium outcomes.

My main results, Theorems 1–3, characterize equilibrium properties of the moral game. Theorem 3 establishes that a heterogeneous outcome can be supported in a Reluctance to Change SPNE only when no homogeneous outcome strictly Pareto dominates it in the material game. The intuition is that if a homogeneous outcome Pareto dominates a heterogeneous one in material terms, the influencing-outcome motive gives at least one player an incentive to redirect moral pressure toward the homogeneous outcome, making the heterogeneous outcome unsupportable. This result identifies a necessary condition for moral diversity. Material payoffs must not uniformly favor conformity. Theorem 2 shows that when moral weights are scaled up sufficiently, any homogeneous outcome can be supported through self-fulfilling moral functions, in which both players praise the coordinated action and assign low value judgments to alternatives. The intuition is that large moral weights amplify the self-righteous motive, making it rational to praise whatever action one plans to take, regardless of material incentives. This generates a social norm force that can support any uniform behavior when moral influence is strong enough. Theorem 1 shows that any homogeneous outcome that is material game Pareto efficient remains moral-game Pareto efficient. Together, these results identify when moral formation promotes efficiency, enforces conformity, and tolerates diversity.

The analysis proceeds across two dimensions of the moral weight structure, the inequal-

ity parameter *Ine* and the individualism parameter *Ind*. Greater inequality concentrates projecting moral weight on one player, amplifying that player’s influencing-outcome motive. Greater individualism means players weight their own value judgments more relative to others’. Negative individualism, which I call collectivism, reverses this. These two dimensions jointly determine which outcomes the moral mechanism can support in a given material game. Section II defines these parameters formally. Because the equilibrium correspondence cannot be characterized analytically for general payoff matrices, I develop a computational method, implemented in a MATLAB application, that maps the complete equilibrium correspondence across the parameter space for specific games.¹

I apply the framework to ten material games in Section III. In the Prisoners Dilemma, both two-sided cooperation and one-sided cooperation are supported under collectivism with equality or under inequality with moderate individualism, without repeated interaction or exogenous fairness norms. In coordination games (Cooper et al., 1990; Van Huyck et al., 1990), the moral mechanism uniquely selects the efficient equilibrium under high inequality or moderate individualism, because a player with concentrated projecting moral weight can commit to value judgments that make the efficient action dominant. In consumption games, the mechanism supports status-seeking behavior, in which goods consumed by influential individuals acquire value that induces others to consume them. In games with unequal influence, the mechanism produces outcomes in which less influential players endorse moralities that favor the powerful player, providing a strategic account of the internalization of values that serve others’ interests.

This paper contributes to four strands of literature. First, it endogenizes the moral standards that models of social preferences (Rabin, 1993; Fehr and Schmidt, 1999; Akerlof and Kranton, 2000) take as fixed parameters, taking as primitive only that agents care about receiving value judgments. Second, unlike repeated-game approaches to cooperation (Friedman, 1971; Fudenberg and Maskin, 1986; Kandori et al., 1993), my mechanism operates in a static two-stage game through the direct payoff consequences of moral functions, without punishment threats or discounting. Third, unlike models of social norms that impose conformity exogenously (Akerlof, 1980; Bernheim, 1994), Theorem 2 derives the norm force endogenously from individual incentives. Fourth, unlike the indirect evolutionary approach (Güth and Yaari, 1992; Güth, 1995; Dekel et al., 2007; Heifetz et al., 2007), moral functions are driven by selfish motives, and not assigned by evolutionary pressure.

¹The algorithm is described in Appendix C.

I Literature Review

This paper relates to several bodies of work, including models of social preferences, information-based models of prosocial behavior, repeated-game and equilibrium selection approaches to cooperation, models of social norms, and the indirect evolutionary approach. In each case, the key difference is that I endogenize the moral standards themselves rather than treating preferences, behavioral prescriptions, or preference types as fixed parameters.

A large literature embeds social components directly into utility functions. [Becker \(1974\)](#) pioneered this approach by incorporating others' wellbeing into individual preferences. Altruism models specify that utility depends directly on others' utilities ([Andreoni, 1989, 1990](#)). Reciprocity and fairness models introduce preferences over both outcomes and intentions ([Rabin, 1993](#); [Fehr and Gächter, 2000](#); [Dufwenberg and Kirchsteiger, 2004](#); [Falk and Fischbacher, 2006](#)). Psychological game theory ([Geanakoplos et al., 1989](#); [Battigalli and Dufwenberg, 2011](#)) specifies utilities that depend on beliefs about others' kindness. Inequity aversion models ([Fehr and Schmidt, 1999](#); [Bolton and Ockenfels, 2000](#)) incorporate preferences over payoff distributions. Identity economics ([Akerlof and Kranton, 2000, 2005](#)) ties utility to identity categories and the behavioral prescriptions associated with them. [Bénabou and Tirole \(2011\)](#) extend this by modeling beliefs about one's own moral identity as durable cognitive assets that individuals actively manage through selective memory and motivated reasoning, with moral behavior driven by the demand for a positive self-image. In all these models, the direction and intensity of social concerns are exogenously specified as preference parameters.

The present model differs in its primitive. Standard social preference models ask how prosocial preferences affect outcomes given fixed moral standards. This paper asks how strategic agents choose moral standards to achieve desired outcomes. What enters utility is not others' material payoffs or the kindness of their intentions, but the value judgments others assign to one's actions. The content of those judgments is endogenously determined by material incentives rather than fixed as a predetermined feature of preferences.

Several specific contrasts follow. In altruism models, the effectiveness of prosocial behavior can diminish as group size grows. In the present model, cooperation is supported through social recognition. Through the influencing-outcome motive, individuals assign high value judgments to actions that benefit themselves, creating incentives for others to take those actions. In fairness models, players inherently care whether others are kind; in the present model, kindness itself is strategically defined, so reciprocal moral judgments emerge from incentives rather than from predetermined fairness criteria. In identity economics, behavioral prescriptions are exogenous features that agents take as given; here, those prescriptions are

the strategic choice variables.

Reputation models (Kreps et al., 1982; Milgrom and Roberts, 1982; Kreps, 1990), costly signaling models (Spence, 1973; Gintis et al., 2001), social image models (Bénabou and Tirole, 2006), and cheap talk models (Crawford and Sobel, 1982; Farrell and Rabin, 1996) all explain prosocial behavior through private information and type uncertainty. Cooperative or prosocial types have characteristics that rational agents may mimic or signal. The present framework assumes no private information. All preferences over material payoffs are common knowledge. The heterogeneity that emerges in what counts as good versus bad behavior arises from strategic choices, not from hidden types to be inferred through signaling or information extraction.

The Folk Theorem (Friedman, 1971; Fudenberg and Maskin, 1986) shows that cooperation can be supported in infinitely repeated games through the threat of reverting to stage-game Nash equilibria. The present mechanism differs in two respects. First, it operates in a two-stage game played once, without discounting or repeated interaction. Second, it can support one-sided outcomes in which one player’s payoff falls below the stage-game Nash equilibrium level, which standard Folk Theorem constructions do not deliver. At the extreme, the mechanism sustains outcomes of forced moral imposition, in which one party endorses a moral standard that serves only the other’s interests, as in institutions of slavery, colonialism, or caste hierarchy. Power shifts between players through adjustments in moral weights rather than through punishment sequences.

Cooper et al. (1990), Van Huyck et al. (1990), and related work study equilibrium selection in coordination games. Pre-play communication (Cooper et al., 1992), focal points (Schelling, 1960), evolutionary dynamics (Maynard Smith, 1982; Kandori et al., 1993; Young, 1993), and global games (Carlsson and Van Damme, 1993; Morris and Shin, 2003) have each been proposed as selection mechanisms. The moral mechanism operates differently. A player with concentrated projecting moral weight commits to value judgments that make the efficient action dominant for the other player, selecting the efficient equilibrium without communication, salience, or perturbed information.

The social norms literature (Akerlof, 1980; Bernheim, 1994; Young, 1993, 1998; Bicchieri, 2006) studies how behavioral regularities emerge and persist through social sanctions and conformity pressures. Most models treat conformity pressure as exogenous, or derive it from evolutionary dynamics (Axelrod, 1986; Boyd and Richerson, 1985; Young, 1998). Theorem 2 derives the norm force endogenously. When moral weights are scaled up sufficiently, any homogeneous outcome can be supported through self-fulfilling moral functions, in which both players praise the coordinated action and assign low value judgments to alternatives. Norm emergence follows from individual incentives rather than from evolutionary selection

or exogenously imposed sanctions.

The indirect evolutionary approach (Güth and Yaari, 1992; Güth, 1995; Dekel et al., 2007; Heifetz et al., 2007) studies whether preferences that deviate from material payoff maximization can survive evolutionary selection. Individuals are randomly matched and play Nash equilibria given their types; types with higher fitness reproduce faster and types with lower fitness die out. Three differences from the present paper stand out. First, moral functions in my model affect everyone’s preferences, not only the player who chooses them. An agent does not choose to care about others but chooses what standard all players apply when evaluating actions. Second, there is no external evolutionary pressure selecting types. Equilibrium is determined by the influencing-outcome and self-righteous motives, not by fitness dynamics. Third, because evolutionary selection eliminates unfit types, the indirect evolutionary approach yields outcomes in which surviving agents earn similar material payoffs. The present paper produces equilibria in which one player adopts a moral function favorable to the other, generating persistent payoff inequality, and these outcomes need not satisfy any fitness-neutrality requirement.

II Model

Let a *material game* be a normal form game with two players 1 and 2 simultaneously choosing action $a_i \in \{A, B\}$ for player $i \in \{1, 2\}$. I denote the *material payoff* for player i from action profile (a_1, a_2) as $u_i^M(a_1, a_2)$.

A *moral game* is a two-stage game built on top of a material game. In stage 1, each player i simultaneously chooses a *moral function* $m_i : \{A, B\} \rightarrow [0, 1]$, where $m_i(a)$ is player i ’s *value judgment* of action a . A value of 1 denotes good and virtuous; a value of 0 denotes bad and vicious.² In stage 2, players observe all chosen moralities and simultaneously select actions. Because moral functions are chosen before actions are taken, they constitute credible commitments rather than ex-post rationalizations. Players have no prior or fixed moral position; they form their moral functions endogenously.

The total payoff for player i is $u_i = u_i^M(a_1, a_2) + u_i^S$, where u_i^S is the *social payoff*.³ Additive separability between material and social payoffs is assumed for analytical tractability. The social payoff equals $u_i^S = \sum_j w_{ij} m_j(a_j)$, where $w_{ij} > 0$ for $i, j \in \{1, 2\}$, a_i is player

²The formulation is not restricted to explicitly moral terminology. Any evaluative term whose connotation enters preferences in this way qualifies. This includes positive terms such as right, honest, kind, and successful, and negative terms such as evil, uncivilized, and unjust.

³To see why this payoff structure is natural, consider a world in which value judgments had no effect on behavior. Individuals would face no social cost from actions others judge bad and no social benefit from actions others judge good. The social payoff u_i^S captures exactly this influence.

i 's stage-2 action, and $m_j(a_i)$ is player j 's value judgment of that action. Since there are only two actions, I write $\alpha_j = m_j(A)$ and $\beta_j = m_j(B)$. The *moral weight* w_{ij} measures how much player i cares about player j 's value judgment of i 's action. The *receiving moral weight* of player i is $\sum_j w_{ij}$, the total weight player i places on all value judgments. The *projecting moral weight* of player j is $\sum_i w_{ij}$, the aggregate weight all players place on player j 's value judgments. I refer to a player as *powerful* or *influential* when that player has greater projecting moral weight.

All moral weights $w_{ij} > 0$ are exogenously given. Positivity rules out the degenerate case where players are indifferent to value judgments.⁴

Table 1 gives the stage-2 subgame payoff matrix. In stage 2, players know all chosen moralities. Actions associated with higher value judgments carry added payoff incentives, so morality functions as a commitment device that alters the effective payoffs in the subgame.

Table 1: Subgame payoff matrix of the moral game		
	A	B
A	$u_1^M(A, A) + w_{11}\alpha_1 + w_{12}\alpha_2$ $u_2^M(A, A) + w_{21}\alpha_1 + w_{22}\alpha_2$	$u_1^M(A, B) + w_{11}\alpha_1 + w_{12}\alpha_2$ $u_2^M(A, B) + w_{21}\beta_1 + w_{22}\beta_2$
B	$u_1^M(B, A) + w_{11}\beta_1 + w_{12}\beta_2$ $u_2^M(B, A) + w_{21}\alpha_1 + w_{22}\alpha_2$	$u_1^M(B, B) + w_{11}\beta_1 + w_{12}\beta_2$ $u_2^M(B, B) + w_{21}\beta_1 + w_{22}\beta_2$

In stage 1, each player foresees the stage-2 subgame outcome when choosing a moral function. Two behavioral forces operate. The *influencing-outcome motive* is the incentive to praise actions that benefit oneself, inducing the other player to take those actions in stage 2. This makes morality a strategic tool for internalizing externalities. The *self-righteous motive* is the incentive to assign a high value judgment to the action one plans to take in stage 2, thereby gaining social payoff from self-assessment. Under an SPNE, a player foresees her own stage-2 action and pre-assigns it a high value judgment. This self-fulfilling structure generates multiplicity of equilibrium outcomes.

Consider a smoker and a non-smoker sharing a public space. The smoker prefers to smoke; the non-smoker prefers to avoid secondhand smoke. The non-smoker can give a disapproving look or make a negative comment on smoking in public. This is a stage-1 moral choice. How much the smoker is affected depends on the moral weight she places on the non-smoker's judgment. If she anticipates smoking in equilibrium, the self-righteous motive leads her to pre-assign a high value judgment to smoking, for example by reasoning that smoking is her

⁴Allowing $w_{ij} \leq 0$ would permit players to be averse to certain value judgments, actively preferring to act against what others approve. This extension is left for future work.

right and that secondhand smoke is none of her concern, thereby increasing her social payoff from self-approval.

This example illustrates the model on a personal level. Individuals interact face-to-face, with moral influence transmitted through words and gestures. The moral weight w_{ij} grows with contact frequency and social proximity. Among family members, colleagues, or regular acquaintances, moral judgments are communicated often, so the moral weight is high. Among strangers, it is low.

The same mechanism extends to larger platforms such as education, media, and religious institutions. In such settings, the moral game can be viewed as a reduced-form representation of a weighted social influence process over shared ethical standards. Consider media. Each outlet frames events with a moral valence. In a market economy, wealthy individuals and interest groups fund outlets that praise actions or policies favorable to themselves, exercising the influencing-outcome motive on the supply side. Ordinary consumers choose outlets that elevate their social standing or validate their own views, exercising the self-righteous motive on the demand side. The moral content a market produces can be approximated by a weighted aggregate of these moral choices, with weights determined by each agent's ability to shape what others receive.

I implement two equilibrium concepts. The first is pure-strategy Subgame Perfect Nash Equilibrium (SPNE). The second is pure-strategy Reluctance to Change SPNE (RTC SPNE).

Definition 1. A pure-strategy SPNE of a moral game consists of

$$\{(\alpha_1^*, \beta_1^*, f_1(m_1, m_2)), (\alpha_2^*, \beta_2^*, f_2(m_1, m_2))\},$$

where $m_i^* = \{\alpha_i^*, \beta_i^*\}$ is the on-path morality of i , and $f_1, f_2 : [0, 1]^4 \rightarrow \{A, B\}^2$ are the pure strategies in response to all possible moralities in the first period. Subgame perfect Nash equilibrium requires that, given the moralities chosen in the first stage, both on- and off-path actions chosen in the second stage best respond to those of the other player.

Definition 2. A pure-strategy Reluctance to Change SPNE (RTC SPNE) is a pure-strategy SPNE

$$\{(\alpha_1^*, \beta_1^*, f_1(m_1, m_2)), (\alpha_2^*, \beta_2^*, f_2(m_1, m_2))\}$$

where $f_i(m_1, m_2) = f_i(m_1^*, m_2^*)$ for all i, m_1, m_2 such that $\{f_1(m_1, m_2), f_2(m_1, m_2)\}$ is a subgame Nash equilibrium for $\{m_1, m_2\}$.

RTC SPNE requires off-path strategies to maintain the same stage-2 Nash equilibrium action pair as the on-path pair, as long as the on-path actions remain feasible Nash equilibria given the deviated moralities. This refinement disciplines off-path behavior when multiple

pure-strategy stage-2 Nash equilibria exist for a given morality pair. Players do not switch between equilibria in response to marginal changes in moralities; they change their actions only when the current Nash equilibrium becomes infeasible. The refinement captures behavioral inertia, an environment in which outcomes are stable and continuous with respect to small perturbations in moral beliefs.

I restrict attention to material games in which each player has exactly two actions, A and B , and both players share the same action set, so that actions can be assigned value judgments uniformly across players.⁵

The moral game models a mechanism in which each agent has the capacity to shift the preferences of others toward or against any action. This is analogous to a mechanism design problem, except that every agent, rather than one, has influencing capacity. The model serves as a reduced-form representation of any decentralized reward and punishment mechanism in which rewards and punishments are direct and explicit and do not interact with the material payoffs of the underlying game. This distinguishes the moral game from repeated-game punishment strategies such as grim trigger, which alter continuation payoffs.

The moral game does not feature a material cost that depends on the moral function chosen or on the value judgments given in equilibrium, but the framework can be extended to incorporate such costs, as would arise in settings involving monetary transfers or legal sanctions that depend on the rewards and punishments actually carried out. The mechanism need not be moral in the literal sense. For moral applications, the cost of issuing a value judgment is negligible, as when disapproval is expressed through words or gestures, as in the smoking example.⁶

The model rests on four conditions. First, agents must have sufficient power to influence each other's behavior. Second, moral functions must be established before the relevant actions are taken. This establishment need not be explicit. In the absence of a formal announcement, others infer the moral functions from equilibrium. Third, moral functions must be credible. Moral doctrines are sticky in practice and change only slowly. People

⁵Extending the analysis to richer action sets is straightforward but substantially complicates the characterization of equilibrium correspondences.

⁶Unlike monetary and legal enforcement mechanisms, which require observation by others to be implemented, the moral mechanism can function when some actions are observed only by the agent herself. Monetary and legal punishments are administered by others and cannot be enforced without observation. The moral function, by contrast, is defined over actions unconditionally on whether others observe them. To see why this matters, consider a richer model in which an agent takes an action twice in stage 2, once in a setting observed by others and once in a setting observed only by herself, as in the difference between stealing in public and stealing in private. When the moral function is defined over actions unconditionally on whether others observe them, the self value judgment applied to the privately observed action carries over the effect of others' value judgments through the equilibrium self-term w_{ii} . The moral mechanism thereby functions even when some actions are private. Monetary and legal enforcement stops where observation stops.

often perceive moral standards as time-independent and universally true, not as strategies that can be freely adjusted. A moral position rapidly reversed signals insincerity and loses influence. The stickiness of moral positions is what makes stage-1 choices effective commitments. Fourth, players apply the same moral standard to themselves as to others. This is a simplifying assumption that affects analysis but can be relaxed.

The moral game captures two central features of morality in practice. The first is that morality shapes behavior. Morality motivates agents toward actions judged good and away from actions judged bad. The social payoff u_i^S formalizes this force. The second is that moral functions are chosen strategically per person. I identify four motives that drive this choice. The influencing-outcome motive and the self-righteous motive are modeled in this paper. The third motive is moral enforcement pressure. Those who attribute moral standards to external sources such as indoctrination are referring to this motive. In a richer model, moral functions could take other moral functions as arguments, so that agents reward those who hold compatible moral functions and punish those who hold deviant ones, creating conformity pressure. I do not model this because it significantly complicates the analysis. My conjecture is that its main effect is to reinforce the social norm force in Theorem 2.⁷ The fourth motive is innate moral preference, an exogenous preference over moral functions themselves, independent of material or social incentives. An innate preference shifts the equilibrium outcome to that particular favored moral function and the corresponding action, and is not strategically interesting. This extension can be incorporated easily into the framework.⁸

The model accommodates contradictory moral standards across societies and historical periods. Slavery was once praised as an institution of social order and is now condemned. In the present model, this shift corresponds to a change in both the material payoff structure and the moral weight structure. Free labor yields more production than slavery as technology progresses, and industrialists, growing richer, gain greater ability to influence social ethics. These forces together shift the equilibrium moralities and actions. The model can simulate historical and social changes in morality by varying the moral-weight network or the material game structure.

⁷A way to extend the model to capture life cycle effects is an overlapping-generations structure. Individuals choose part of their morality in youth, retain those values in adulthood, and acquire additional moral values later in life. Children primarily have receiving weights and are thus more susceptible to parental influence, while adults acquire more projecting weights and greater capacity to influence others. This structure complements cultural transmission models (Boyd and Richerson, 1985; Bisin and Verdier, 2001) by endogenizing the strategic formation of moral values across life stages.

⁸In practice, innate moral preferences are likely scarce. A truly innate moral preference must exclude any of the previous three motives, or equivalently, be adopted without influence from other people, live or dead, and without serving any material or self-interest.

III Application

I apply the moral game framework to ten material games. Given the exogenous parameters $u_i^M(a_1, a_2)$ for all action profiles and w_{ji} for all i, j , I characterize the action pairs supported as on-path outcomes. Once the supported action pairs are identified, the corresponding moral functions are easy to describe; the substantive task is identifying which action pairs can be supported. A pair of subgame actions $(a_1, a_2) \in \{A, B\}^2$ is **supported** by the parameters when there exist $\alpha_1^*, \alpha_2^*, \beta_1^*, \beta_2^* \in [0, 1]$ such that $\{(\alpha_1^*, \beta_1^*, f_1(m_1, m_2)), (\alpha_2^*, \beta_2^*, f_2(m_1, m_2))\}$ is an SPNE, and $a_1 = f_1(\alpha_1^*, \beta_1^*, \alpha_2^*, \beta_2^*)$, $a_2 = f_2(\alpha_1^*, \beta_1^*, \alpha_2^*, \beta_2^*)$ are the on-path actions.

The first two applications establish general theoretical results about value judgments, efficiency, and social norms. Subsequent applications illustrate how these forces interact with specific strategic environments.

To characterize and visualize results, I reduce the four moral-weight parameters to three by introducing total weight, inequality, and individualism. Total weight is $W = \sum_{i,j} w_{ji}$. The inequality parameter is Ine and the individualism parameter is Ind . Moral weights are

$$\begin{aligned} w_{11} &= W \cdot \left(\frac{2^{Ine}}{2^{Ine} + 1} \right) \cdot \left(\frac{2^{Ind}}{2^{Ind} + 1} \right) \\ w_{21} &= W \cdot \left(\frac{2^{Ine}}{2^{Ine} + 1} \right) \cdot \left(\frac{1}{2^{Ind} + 1} \right) \\ w_{12} &= W \cdot \left(\frac{1}{2^{Ine} + 1} \right) \cdot \left(\frac{1}{2^{Ind} + 1} \right) \\ w_{22} &= W \cdot \left(\frac{1}{2^{Ine} + 1} \right) \cdot \left(\frac{2^{Ind}}{2^{Ind} + 1} \right). \end{aligned}$$

I adopt this reduction primarily for visualization, since four-dimensional parameter spaces are difficult to display.^{9,10} The inequality parameter Ine determines the relative projecting moral weight of the two players. Greater inequality gives player 1 more projecting moral weight, $\sum_i w_{i1}$. At $Ine = 0$, the two players have equal projecting moral weights. Negative values of Ine correspond to player 2 having more projecting moral weight. The individualism parameter Ind determines how much players weight their own value judgments relative to others'. Greater individualism means players care more about their own value judgment. At $Ind = 0$, both players weight self and other equally, $w_{i1} = w_{i2}$ for all i . I refer to negative

⁹The exponential functional form yields interpretable dimensions. Alternative parameterizations are possible but do not change the qualitative results.

¹⁰This reduction imposes a common individualism level across agents. The assumption can be violated under a Maslowian preference hierarchy, where middle-income agents place greater weight on social recognition while agents facing subsistence constraints prioritize basic needs. Allowing agent-specific individualism parameters would expand the dimensionality of the parameter space and could reveal richer patterns of equilibrium selection across socioeconomic strata.

individualism as *collectivism*, since players then care more about others' value judgments than about their own.

A MATLAB App Designer application visualizes supported outcomes under SPNE and RTC SPNE and accepts user-specified material payoffs. Each subgame outcome receives a numerical code, with (A, A) assigned 1, (A, B) assigned 2, (B, A) assigned 4, and (B, B) assigned 8. A displayed value is the additive sum of the codes for all supported outcomes, yielding a unique label for each supported set. A value of 9 means the parameters support both (A, A) and (B, B) ; a value of 15 means all four outcomes are supported. In the figures below, the horizontal axis represents *Ine* and the vertical axis represents *Ind*. Each colored region corresponds to a set of supported outcomes. Values 1, 2, 4, or 8 indicate unique equilibrium selection; additive values such as 3, 9, or 15 indicate multiplicity. The left panel in each figure uses lower total moral weight W and the right panel uses higher W , illustrating how stronger moral influence affects equilibrium selection. The SPNE and RTC SPNE equilibrium correspondences are identical for all games in this paper except the Inverse Battle of the Sexes at $W = 8$, for which the two graphs are shown separately in Figure 13.

A Indifference Moral Game

The indifference moral game sets all material payoffs to zero (Table 2).

Table 2: Payoff matrix for the indifference game

	A	B
A	0,0	0,0
B	0,0	0,0

Since all material payoffs are zero, the stage-2 subgame payoff matrix reduces to Table 3.

Table 3: Subgame payoff matrix for the indifference moral game

	A	B
A	$w_{11}\alpha_1 + w_{12}\alpha_2,$ $w_{21}\alpha_1 + w_{22}\alpha_2$	$w_{11}\alpha_1 + w_{12}\alpha_2,$ $w_{21}\beta_1 + w_{22}\beta_2$
B	$w_{11}\beta_1 + w_{12}\beta_2,$ $w_{21}\alpha_1 + w_{22}\alpha_2$	$w_{11}\beta_1 + w_{12}\beta_2,$ $w_{21}\beta_1 + w_{22}\beta_2$

The indifference moral game isolates the self-righteous motive. Because all material payoffs are zero, the influencing-outcome motive is absent. Other moral games share this self-righteous structure but add a trade-off between the two motives.

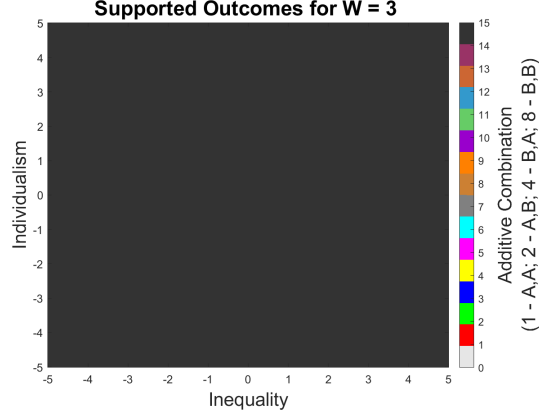


Figure 1: Supported subgame outcomes in the indifference moral game ($W = 3$)

Definition 3. A pair of actions is *homogeneous* if they are the same.

Definition 4. A pair of actions is *heterogeneous* if they are distinct.

Lemma 1. *For all moral games, any RTC SPNE*

$$\{(\alpha_1^*, \beta_1^*, f_1(m_1, m_2)), (\alpha_2^*, \beta_2^*, f_2(m_1, m_2))\},$$

satisfies the following. There exists $\gamma \in \{\alpha_i, \beta_i\}$ such that $\gamma = 1$ for any player i . Moreover, if the on-path action pair is homogeneous in some RTC SPNE, then it must have $\alpha_1^ = \alpha_2^* = 1$ if the on-path action pair is (A, A) and $\beta_1^* = \beta_2^* = 1$ otherwise.*

(Proof.)

The intuition is that only the difference $\alpha_i - \beta_i$ determines each player's stage-2 incentives. Holding that difference fixed, every player strictly prefers to raise their value judgments to the upper bound, since doing so increases social payoff without affecting the subgame outcome. For homogeneous action profiles, this implies that both players assign value judgment 1 to the on-path action.

Definition 5. An action pair (a_1, a_2) *strictly Pareto dominates* (a'_1, a'_2) in the material game if $u_i^M(a_1, a_2) > u_i^M(a'_1, a'_2)$ for all i ; it *weakly Pareto dominates* (a'_1, a'_2) if $u_i^M(a_1, a_2) \geq u_i^M(a'_1, a'_2)$ for all i and $u_i^M(a_1, a_2) > u_i^M(a'_1, a'_2)$ for some i . An action pair is *material game Pareto efficient* if no alternative action pair weakly Pareto dominates it in the material game. An RTC SPNE $\{(\alpha_1^*, \beta_1^*, f_1), (\alpha_2^*, \beta_2^*, f_2)\}$ *strictly Pareto dominates* another RTC SPNE if it gives every player a strictly higher total payoff $u_i = u_i^M + u_i^S$; it *weakly Pareto dominates* another RTC SPNE if it gives every player at least as high a total payoff and at least one player a strictly higher total payoff. An RTC SPNE is *moral-game Pareto efficient* if no other RTC SPNE weakly Pareto dominates it.

Theorem 1. *Any RTC SPNE that supports an on-path homogeneous subgame outcome that is material game Pareto efficient is moral-game Pareto efficient.*

(*Proof.*)

The intuition behind Theorem 1 is that when the on-path action pair is homogeneous, Lemma 1 guarantees that social payoff is maximized. A homogeneous outcome that is material game Pareto efficient therefore achieves the maximum possible social payoff as well, making it moral-game Pareto efficient. The same argument does not apply to heterogeneous outcomes, which generally forgo social-payoff gains.

Proposition 1. *All subgame outcomes are supported under the indifference moral game.*

(*Proof.*)

The intuition behind Proposition 1 is that, in the indifference moral game, constructing a supporting morality pair reduces to checking whether each outcome can be a stage-2 Nash equilibrium. For homogeneous outcomes, setting $\alpha_i - \beta_i = 1$ or $\alpha_i - \beta_i = -1$ for both i suffices, and this is always achievable when all material payoffs are zero. For heterogeneous outcomes, the test is more demanding. Players choose different actions, and praising one action benefits one player while hurting the other. Nevertheless, with all material payoffs equal to zero, the moral incentives alone are sufficient to support any outcome, including heterogeneous ones.

Theorem 2 (Social Norm). *For any moral game, there exists $k' \in \mathbb{R}_+$ such that, for all $k \geq k'$, under the new moral game with the same material game payoffs but new weights $w'_{ij} = kw_{ij}$ for all i, j , both homogeneous subgame outcomes (A, A) and (B, B) can be supported as RTC SPNE.*

(*Proof.*)

Corollary 1. *For any moral game that contains a homogeneous subgame outcome that is material game Pareto efficient, there exists $k' \in \mathbb{R}_+$ such that, under the new moral game with the same material game payoffs but new weights $w'_{ij} = kw_{ij}$ for all i, j and $k \geq k'$, there exists an RTC SPNE that is moral-game Pareto efficient.*

(*Proof.*)

Theorem 2 shows that large moral weights generate a social norm force that can support any homogeneous outcome. The mechanism is self-fulfilling, in that both players praise the coordinated action and assign low value judgments to the alternative, making the coordinated action a dominant choice in stage 2. The same logic does not apply to heterogeneous outcomes. Subsequent applications exhibit material games in which no heterogeneous outcome is supportable at any weight level.

Proposition 2. *An indifference moral game RTC SPNE is always Pareto efficient.*

(*Proof.*)

The intuition behind Proposition 2 is that without the influencing-outcome motive, every player strictly prefers to praise the action they are taking, so any equilibrium morality assigns value judgment 1 to the on-path action. By Lemma 1, this implies maximum social payoff, and Pareto efficiency follows. Any heterogeneous outcome supported in the indifference moral game is also fragile, in that a small perturbation of either player’s morality causes the other player to best respond to homogeneous actions.¹¹

B Private Goods with Similar Preference Moral Game

The PGSP material game is non-strategic and has no externalities. This corresponds to the classical private goods consumption environment, in which one’s own consumption does not enter another’s utility function. The PGSP material game payoff matrix is shown in Table 4.

Table 4: Payoff matrix for the Private Goods with Similar Preference game

	A	B
A	1,1	1,0
B	0,1	0,0

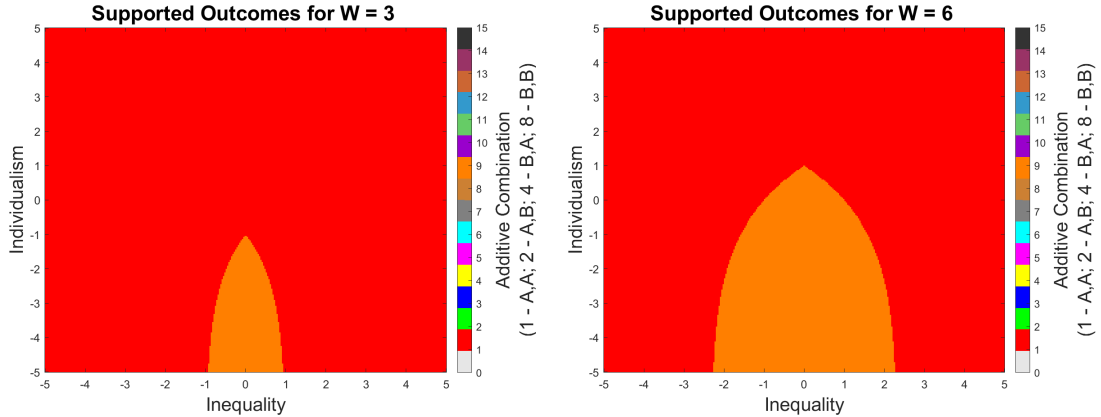


Figure 2: Supported subgame outcomes in the PGSP moral game ($W = 3, 6$ left to right)

Figure 2 shows the supported subgame outcomes across moral-weight parameters. The region coded as 1 supports only (A, A) . The region coded as $9 (= 1 + 8)$ supports both

¹¹An alternative formulation in which players care about value judgments relative to others rather than in absolute terms weakens the social norm force. In such a setting, players have an incentive to push for heterogeneous outcomes to elevate their own relative standing, reducing the pull toward homogeneous coordination.

(A, A) and (B, B) . As moral weights increase, the parameter space supporting the inefficient outcome (B, B) expands, since stronger moral influence allows the social norm force to support the dominated outcome.

Outcome (B, B) is supported when both players set $\alpha_i - \beta_i$ sufficiently negative. Once in this equilibrium, players remain at the inefficient outcome while assigning high value judgments to it, so that the social-payoff loss from deviating to the efficient action offsets any material gain.

In the PGSP moral game, heterogeneous outcomes cannot occur in any RTC SPNE. The intuition is that supporting a heterogeneous outcome requires equilibrium moralities with low $|\alpha_i - \beta_i|$ for both players, whereas homogeneous outcomes can attain $|\alpha_i - \beta_i| = 1$. The higher value-judgment differential available to homogeneous outcomes makes them more resistant to unilateral deviations.

Theorem 3 (Material Efficiency Requirement under RTC SPNE). *For any heterogeneous outcome (i.e., (A, B) or (B, A)) to be supported in a pure-strategy RTC SPNE, neither heterogeneous outcome can be strictly Pareto dominated by any homogeneous outcome (i.e., (A, A) or (B, B)) in the material game.*

(*Proof.*)

The intuition behind Theorem 3 is that heterogeneous outcomes cannot rely on the self-fulfilling mechanism that supports homogeneous ones. By Lemma 1, any homogeneous outcome supported in an RTC SPNE attains the maximum social payoff available to both players, whereas heterogeneous outcomes generally forgo this social-payoff gain. A heterogeneous outcome can therefore be supported only when no homogeneous outcome strictly Pareto dominates it in the material game, so that the material-payoff advantage of diversity offsets the social-payoff cost of disagreement for at least one player.

C Private Goods with Different Preference Moral Game

The PGDP material game reflects diverse preferences for private goods consumption. The payoff matrix is shown in Table 5.

Table 5: Payoff matrix for the Private Goods with Different Preference game

	A	B
A	1,0	1,1
B	0,0	0,1

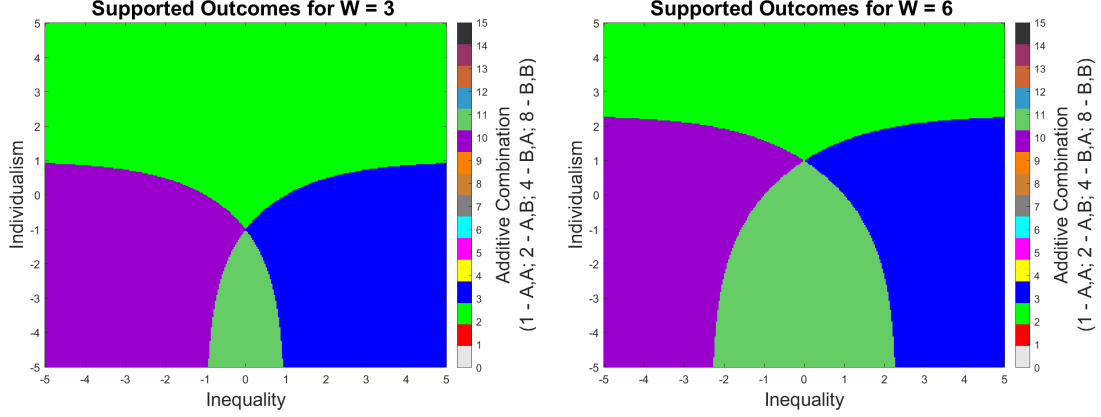


Figure 3: Supported subgame outcomes in the PGDP moral game ($W = 3, 6$ left to right)

The PGDP game illustrates private consumption with heterogeneous preferences, in which players' choices bear no externalities on each other's material utilities but they hold distinct preferences over private goods. For example, richer communities may prefer higher-quality bundles labeled as action A , while poorer communities may prefer lower-quality goods labeled as action B .¹²

Figure 3 shows the supported subgame outcomes. Under high individualism, the on-path outcome of the moral game coincides with the material game Nash equilibrium. Under low individualism with equality, the supported outcomes are either a homogeneous outcome or the material game efficient heterogeneous outcome (A, B) , but never the inefficient heterogeneous outcome (B, A) . Under low individualism with high inequality, the supported outcomes are either (A, B) or the homogeneous outcome preferred by the more influential player, namely (A, A) when player 1 has greater projecting moral weight and (B, B) when player 2 does.

When inequality is large, a heterogeneous outcome can be supported as long as the more influential player refrains from assigning low value judgments to the less-preferred action. This game demonstrates the endogenous rise of conspicuous consumption, in which goods consumed by socially influential players carry additional status that increases others' willingness to consume them.

¹²The PGSP and PGDP games have the same payoffs up to relabeling of actions, yet produce very different equilibrium correspondences. Since moralities are built on whether action pairs are homogeneous or heterogeneous, the labeling of actions matters. Actions should be labeled according to easy and universal distinguishability, whether by the action itself or by identifiable characteristics of the players involved. Labelings that only a few players can apply lack the universality common in contested moral debates.

D Feeding Others Moral Game

The Feeding Others moral game is inspired by the Allegory of the Long Spoons, in which individuals can only feed others, not themselves. Action A corresponds to feeding the other player, and action B to withholding. The payoff matrix is shown in Table 6.

Table 6: Payoff matrix for the Feeding Others game

	A	B
A	1,1	0,1
B	1,0	0,0

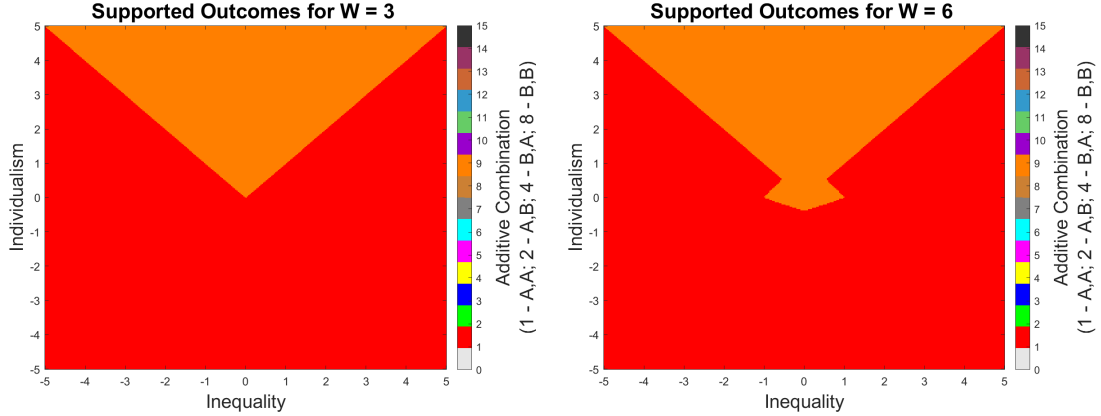


Figure 4: Supported subgame outcomes in the Feeding Others moral game ($W = 3, 6$ left to right)

Figure 4 shows the supported subgame outcomes. Greater collectivism selects the unique efficient outcome (A, A) , and larger absolute inequality does so as well. Under high individualism with equality, both homogeneous subgame outcomes are supported.

The Feeding Others moral game illustrates the influencing-outcome motive directly. Each player benefits from the other's cooperation, so the influencing-outcome motive leads both players to praise action A . When individualism is high, however, this motive weakens because players place less weight on others' value judgments, driving on-path outcomes toward the material game Nash equilibrium. This together with Theorem 3 explains why heterogeneous outcomes are ruled out here. Since (A, A) weakly Pareto dominates both (A, B) and (B, A) in the material game, no heterogeneous outcome can be supported in any RTC SPNE.¹³

¹³Theorem 3 requires strict Pareto dominance for the ruling-out result to hold in general. Here the dominance is only weak, but the same forces apply.

E Personal Favor Moral Game

The Personal Favor material game captures a transfer decision in which action A corresponds to donating to Player 1 and action B to keeping for oneself. The payoff matrix is shown in Table 7.

Table 7: Payoff matrix for the Personal Favor game

	A	B
A	2, 0	1, 1
B	2, 0	1, 1

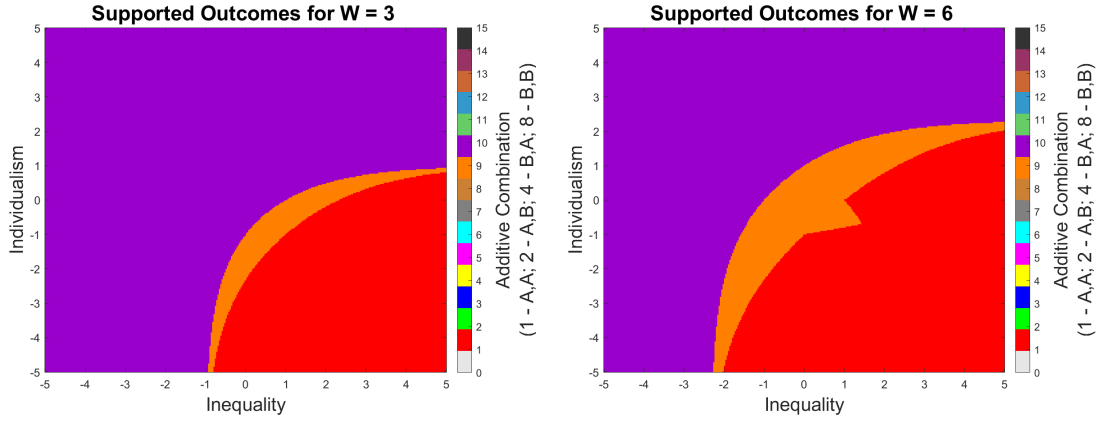


Figure 5: Supported subgame outcomes in the Personal Favor moral game ($W = 3, 6$ left to right)

Player 2 prefers action B in the material game, while Player 1 benefits when Player 2 chooses action A .

Figure 5 shows the supported subgame outcomes across moral-weight parameters. Under collectivism and high inequality favoring Player 1, the outcome (A, A) is supported; by Theorem 1, any such RTC SPNE is moral-game Pareto efficient. When inequality favors Player 2, the supported outcomes are (A, B) and (B, B) , with Player 2 not contributing in either case, because Player 1 lacks sufficient influence to change Player 2's action. When both (A, B) and (B, B) are simultaneously supported, equilibria with on-path outcome (A, B) are weakly Pareto dominated by those with on-path outcome (B, B) . Near equality with moderate projecting weight for Player 1, social norms select only homogeneous outcomes.

The Personal Favor moral game illustrates how rational value internalization can arise from asymmetric influence. When Player 1's projecting moral weight is large and Player 2's is too small to alter the material outcome, Player 2 rationally assigns a high value judgment

to action A despite bearing its material cost. The self-righteous motive then reinforces this assignment, as Player 2 maximizes total payoff by praising the action it will take in any case. Deviating in moral judgments either cannot shift the subgame equilibrium or entails a loss of social recognition from Player 1 that outweighs any material gain. This together with Theorem 1 implies that the resulting homogeneous outcome (A, A) , when it is supported, is moral-game Pareto efficient. In a broader societal context, this mechanism explains how ethical norms that benefit powerful groups can emerge not through coercion but through the rational calculus of the less influential, who optimize within constraints they cannot unilaterally alter.

F Prisoners Dilemma Moral Game

The Prisoners Dilemma material game is the canonical social dilemma in which action A corresponds to cooperation and action B to defection. The payoff matrix is shown in Table 8.

Table 8: Payoff matrix for the Prisoners Dilemma game

	A	B
A	2,2	0,3
B	3,0	1,1

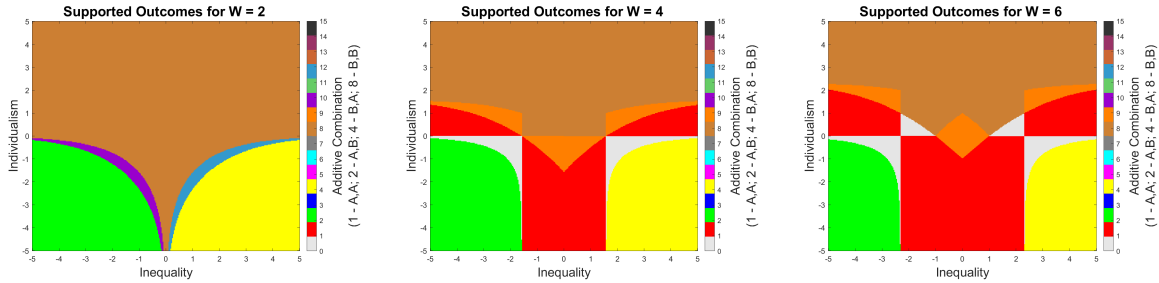


Figure 6: Supported subgame outcomes in the Prisoners Dilemma moral game ($W = 2, 4, 6$ left to right)

The Prisoners Dilemma moral game can be viewed as a Feeding Others game with a cost, or equivalently as a two-sided Personal Favor moral game. Figure 6 shows the supported subgame outcomes across moral-weight parameters. High individualism leads to the material game Nash equilibrium (B, B) . Collectivism and inequality select material game efficient outcomes that favor the more influential player. Under equality with collectivism, and under inequality with moderate individualism, the collaborative egalitarian outcome (A, A) is supported. By Theorem 2, as moral weights increase, homogeneous social-norm outcomes are supported.

The interplay of collectivism with equality and individualism with inequality shapes equilibrium selection. Under equality with collectivism, the collaborative outcome arises because each player benefits from others' cooperation and places weight on others' value judgments, so both players can induce cooperative actions via the influencing-outcome motive. With moderate individualism and inequality, the more influential player must first choose the cooperative action to credibly induce the other to follow, given the weight placed on self-assigned value judgments. This mechanism relies on the influential player adhering to the moral guidelines they advocate for others, and it may fail if the influential player relabels actions, as in the Personal Favor moral game.

The Prisoners Dilemma moral game explains cooperation in cohesive societies without relying on repeated interactions, altruistic preferences, or predefined fairness norms. The moral game mechanism differs from repeated-game Folk Theorem constructions in three ways. First, it supports one-sided cooperative outcomes where one player's payoff falls strictly below their stage-game Nash equilibrium payoff, outcomes that cannot be supported through standard punishment threats in repeated games since the disadvantaged player would prefer reverting to Nash. Second, the moral game permits persistent inequality without requiring alternation between outcomes to divide surplus, as players maintain stable unequal positions rather than cycling through states as in typical Folk Theorem equilibria. Third, shifts in influence between players can be achieved by adjusting moral weights rather than changing material payoffs or invoking reduced-form bargaining protocols. In effect, all agents act as mechanism designers for the other player, though some have more voice.

Unlike the altruism literature, the moral game mechanism can support cooperation as more players join. Denote the all-cooperate action profile by a^C and the profile where player i defects while all others cooperate by a^{D_i} . Two conditions are sufficient to support cooperation. In stage 2, the condition $\sum_j w_{ij} \geq u_i^M(a^{D_i}) - u_i^M(a^C)$ for all i ensures that aggregate moral incentives make cooperation a best response. In stage 1, the condition $\sum_{j \neq i} w_{ij} \geq u_i^M(a^{D_i}) - u_i^M(a^C)$ for all i prevents profitable deviations where a player unilaterally defects while self-assigning high value judgments despite condemnation from others. Neither condition negatively depends on the number of players. As long as each player's receiving moral weight from others remains at least as large as their temptation to defect as the population grows, the cooperative outcome can be supported. However, as the number of players increases, the social norm force also becomes stronger, as each individual's choice of morality has diminishing impact on aggregate value judgments, making unique selection of the cooperative outcome more difficult even when it remains supported.

G Coordination Moral Game

The Coordination material game features two homogeneous Nash equilibria, (A, A) and (B, B) , of which (A, A) strictly Pareto dominates. The payoff matrix is shown in Table 9.

Table 9: Payoff matrix for the Coordination game

	A	B
A	3, 3	0, 0
B	0, 0	1, 1

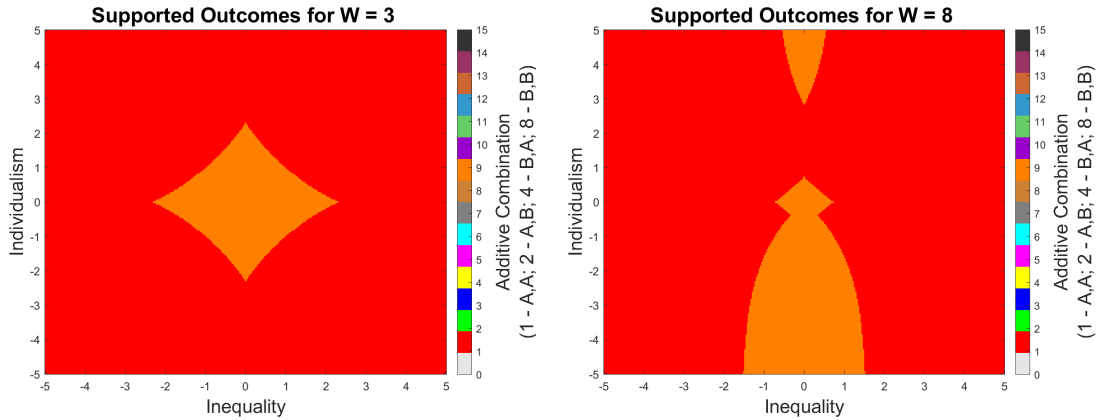


Figure 7: Supported subgame outcomes in the Coordination moral game ($W = 3, 8$ left to right)

Figure 7 shows the supported subgame outcomes across moral-weight parameters. Under high inequality or moderate individualism, the efficient equilibrium (A, A) is uniquely supported, and by Theorem 1, any such RTC SPNE is moral-game Pareto efficient. This configuration overcomes the social norm force that would otherwise lock players into the Pareto-dominated homogeneous outcome (B, B) . The intuition is that a player with sufficient projecting moral weight relative to receiving moral weight can use the influencing-outcome motive to steer the other player toward (A, A) , without being drawn toward the inferior homogeneous outcome by social-payoff considerations. Appropriate moral weights thus resolve the coordination problem by enabling selection of the Pareto-dominant equilibrium through the moral mechanism.

H Inverse Coordination Moral Game

The Inverse Coordination material game is a specialization problem in which players must occupy different roles or positions to achieve efficiency, so action A and action B represent two distinct specializations. The payoff matrix is shown in Table 10.

Table 10: Payoff matrix for the Inverse Coordination game

	A	B
A	0, 0	3, 3
B	1, 1	0, 0

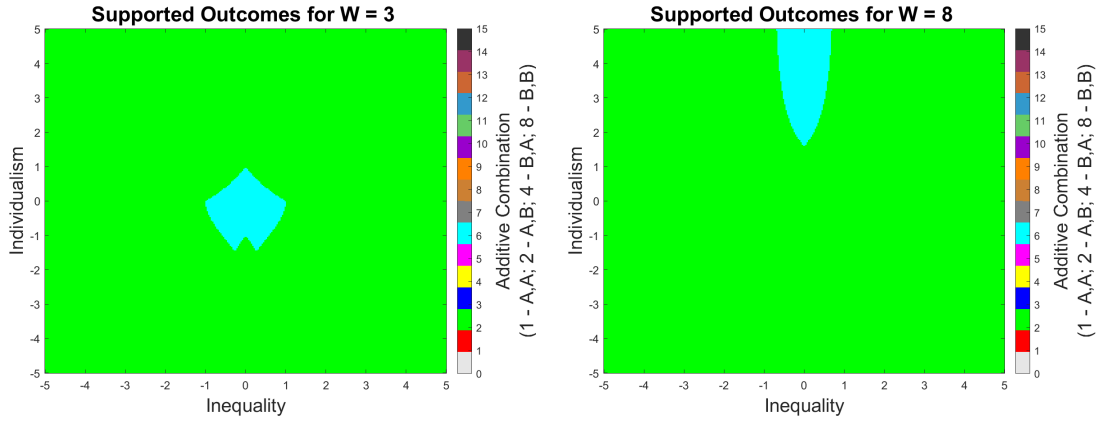


Figure 8: Supported subgame outcomes in the Inverse Coordination moral game ($W = 3, 8$ left to right)

Figure 8 shows the supported subgame outcomes across moral-weight parameters. The Inverse Coordination moral game more reliably supports the efficient heterogeneous outcome (A, B) than the Coordination moral game supports its efficient homogeneous outcome, because here the social norm force and efficiency point in opposite directions. The social norm force promotes homogeneous outcomes, while efficiency requires heterogeneous actions, so the inefficient outcomes are avoided more frequently. At very high total moral weights, when W exceeds approximately 15, the social norm force dominates and both homogeneous outcomes are supported alongside the efficient heterogeneous outcome.

I Stag Hunt Moral Game

The Stag Hunt material game features a payoff-dominant equilibrium (A, A) and a risk-dominant equilibrium (B, B) . Action A corresponds to hunting the stag and action B to

hunting the hare. The payoff matrix is shown in Table 11.

Table 11: Payoff matrix for the Stag Hunt game

	A	B
A	10, 10	1, 8
B	8, 1	5, 5

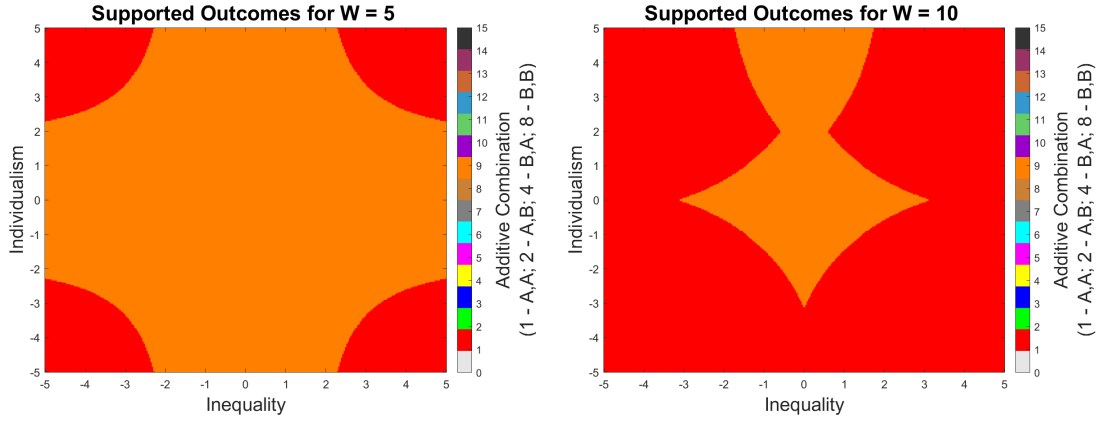


Figure 9: Supported subgame outcomes in the Stag Hunt moral game ($W = 5, 10$ left to right)

Figure 9 shows the supported subgame outcomes across moral-weight parameters. The payoff-dominant equilibrium (A, A) is uniquely supported when either inequality or collectivism is sufficiently high. The intuition behind these two parameter regions differs. Under high inequality, a player with large projecting moral weight and low receiving moral weight can unilaterally implement the efficient equilibrium by using the influencing-outcome motive to induce the other player's cooperation, regardless of social-payoff comparisons. Under high collectivism, both players have strong incentives to encourage cooperation, and the greater weight each places on others' value judgments amplifies each player's ability to induce coordination on (A, A) . In intermediate parameter regions, both the payoff-dominant and risk-dominant equilibria may be simultaneously supported.

J Battle of the Sexes Moral Game

The Battle of the Sexes material game has two efficient homogeneous equilibria, (A, A) and (B, B) , with conflicting preferences in that Player 1 prefers (A, A) and Player 2 prefers (B, B) . The payoff matrix is shown in Table 12.

Table 12: Payoff matrix for the Battle of the Sexes game

	A	B
A	3, 2	0, 0
B	0, 0	2, 3

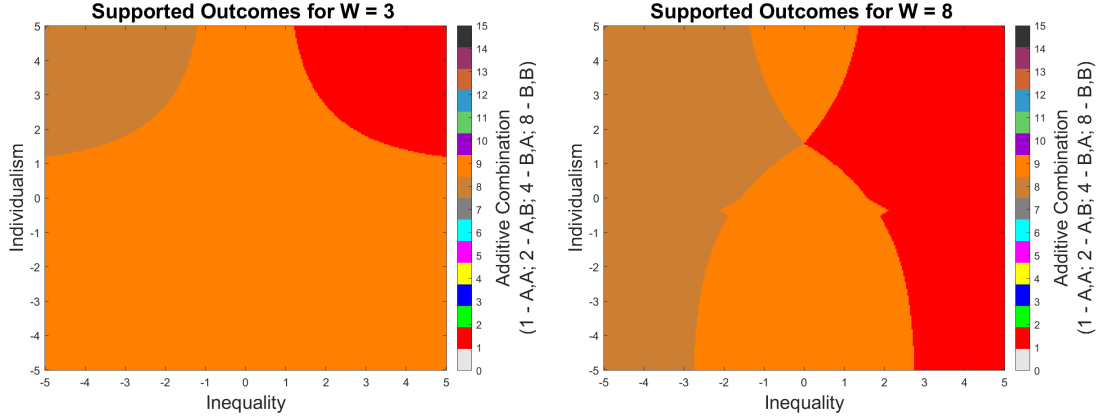


Figure 10: Supported subgame outcomes in the Battle of the Sexes moral game ($W = 3, 8$ left to right)

Figure 10 shows the supported subgame outcomes across moral-weight parameters. Both efficient equilibria (A, A) and (B, B) are supported when inequality is low. High collectivism permits greater inequality while maintaining both efficient equilibria relative to high individualism. The intuition is that individualism and collectivism engage different equilibrium selection forces. Under high individualism, equilibrium stability depends on preventing unilateral deviations in stage 2, where material payoff differences across adjacent outcomes are large relative to social payoffs. Under high collectivism, equilibrium selection is governed by first-stage comparisons between the two homogeneous outcomes, where material payoff differences are modest and social payoffs play a larger role. Collectivist parameters therefore tolerate greater inequality while still supporting both efficient outcomes.

Additional games, including the Chicken moral game and inverse action labelings of Prisoners Dilemma, Stag Hunt, Chicken, and Battle of the Sexes, as well as zero-sum moral games, are analyzed in Appendix A.

IV Discussion

A Empirical Testing

Empirical implementation of the moral game framework requires measuring moral weights and identifying moral functions. Actions are directly observable in many contexts, but moral weights representing the strength of social influence are not. Potential proxies include measures of social proximity such as geographic distance, frequency of interaction, kinship ties, and network centrality in social or professional networks. Digital traces from social media platforms offer measures of attention and influence. Surveys can elicit moral functions through asking participants to evaluate hypothetical or observed actions.

When proxies for moral weights are absent, identification presents a fundamental challenge because multiple configurations of moral weights can support the same observed action profile in equilibrium. One approach is to estimate a moral-weight network from behavior in one strategic environment and test its out-of-sample predictive power in a different game. For instance, moral weights inferred from a dictator game or public goods game can be used to predict behavior in a coordination game, and the model is rejected if the inferred weights systematically fail to predict behavior in the alternative environment.

A second identification strategy exploits exogenous variation in moral weight structure. Natural experiments that alter social influence networks, such as migration, social media platform introductions, or institutional changes affecting hierarchy, provide identifying variation. Panel data tracking the same individuals across different strategic contexts enables within-subject identification of moral weight parameters while controlling for individual fixed effects in moral functions.

APPENDIX

A Additional Games

This appendix presents additional applications of the moral game framework, including the Chicken moral game and inverse action labelings of Prisoners Dilemma, Stag Hunt, Chicken, and Battle of the Sexes, as well as zero-sum moral games. These examples further illustrate the breadth of equilibrium outcomes supported across different moral weight configurations and material game structures.

A Inverse Prisoners Dilemma

Table 13: Payoff matrix for the Inverse Prisoners Dilemma game

	A	B
A	0,3	2,2
B	1,1	3,0

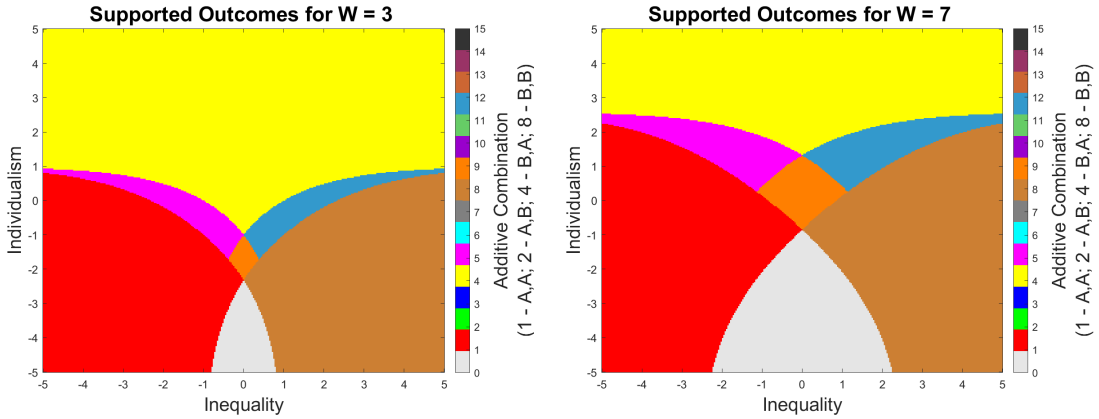


Figure 11: Supported subgame outcomes in the Inverse Prisoners Dilemma moral game ($W = 3, 7$ left to right)

Figure 11 shows the supported subgame outcomes across moral-weight parameters. The Inverse Prisoners Dilemma material game reverses Player 2's action labels relative to the Prisoners Dilemma, so that Player 1's action A still corresponds to cooperation and action B to defection, while for Player 2, action A now corresponds to defection and action B to cooperation; the unique Nash equilibrium of the material game is (B, A) . In contrast

to the Prisoners Dilemma moral game, the Inverse Prisoners Dilemma moral game does not support two-sided cooperation as an equilibrium outcome. Driven by the social norm force, homogeneous outcomes are supported when both the individualism and inequality parameters approach zero. High individualism induces convergence to the material game Nash equilibrium, while high inequality implements the efficient outcome favoring the more influential player.

B Inverse Stag Hunt

Table 14: Payoff matrix for the Inverse Stag Hunt game

	A	B
A	1, 8	10, 10
B	5, 5	8, 1

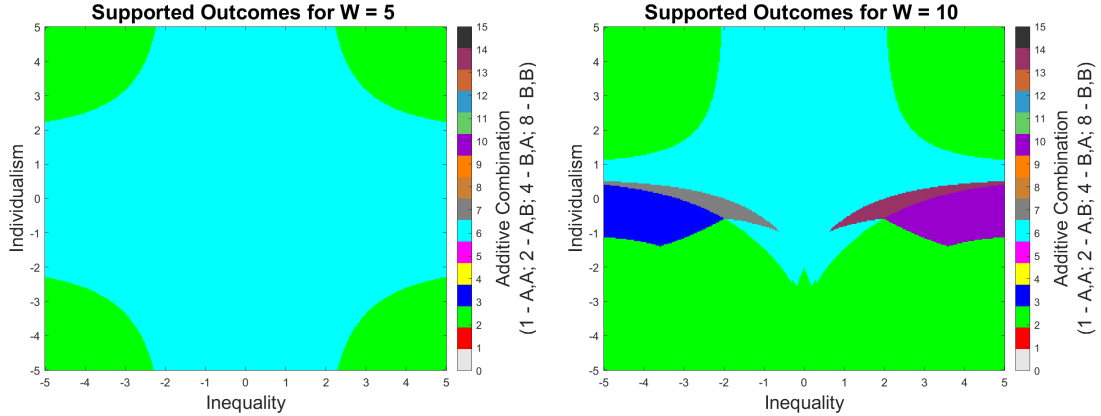


Figure 12: Supported subgame outcomes in the Inverse Stag Hunt moral game ($W = 5, 10$ left to right)

Figure 12 shows the supported subgame outcomes across moral-weight parameters. The Inverse Stag Hunt material game reverses Player 2's action labels relative to the Stag Hunt, so that Player 1's action A still corresponds to hunting the stag and action B to hunting the hare, while for Player 2, action A now corresponds to hunting the hare and action B to hunting the stag; the payoff-dominant equilibrium is (A, B) and the risk-dominant equilibrium is (B, A) . The Inverse Stag Hunt moral game uniquely supports the payoff-dominant equilibrium (A, B) under high collectivism or under high inequality, and this unique selection is maintained as total moral weight increases up until the social norm force enters.

C Inverse Battle of the Sexes

Table 15: Payoff matrix for the Inverse Battle of the Sexes game

	A	B
A	0, 0	3, 2
B	2, 3	0, 0

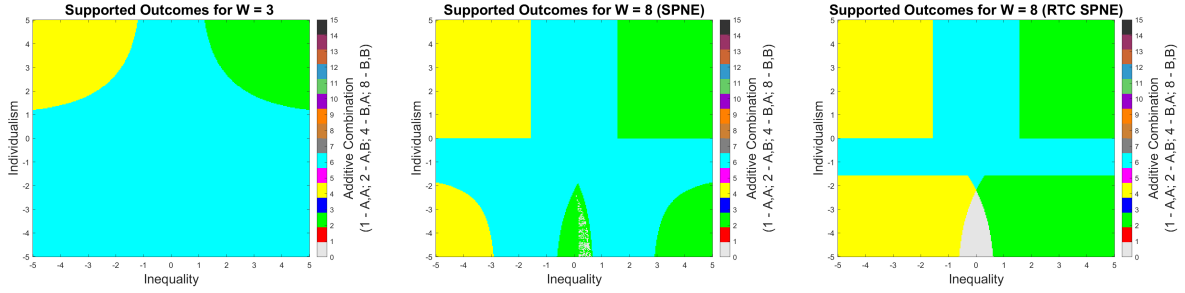


Figure 13: Supported subgame outcomes in the Inverse Battle of the Sexes moral game. Left: SPNE/RTC SPNE, $W = 3$. Centre: SPNE, $W = 8$. Right: RTC SPNE, $W = 8$.

Figure 13 shows the supported subgame outcomes across moral-weight parameters.¹⁴ The Inverse Battle of the Sexes material game reverses Player 2's action labels relative to the Battle of the Sexes. The two Nash equilibria are the heterogeneous outcomes (A, B) and (B, A) . The Inverse Battle of the Sexes moral game exhibits equilibrium patterns qualitatively similar to those of the standard Battle of the Sexes moral game. SPNE as opposed to RTC SPNE supports additional heterogeneous outcomes under high collectivism and moderate inequality.

D Chicken Moral Game

Table 16: Payoff matrix for the Chicken game

	A	B
A	0, 0	-1, 1
B	1, -1	-10, -10

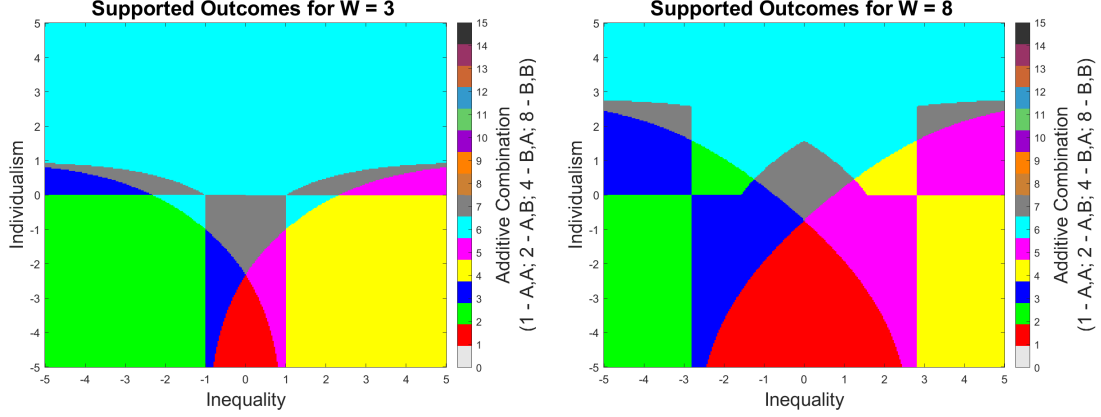


Figure 14: Supported subgame outcomes in the Chicken moral game ($W = 3, 8$ left to right)

Figure 14 shows the supported subgame outcomes across moral-weight parameters. The Chicken material game has two heterogeneous Nash equilibria, (A, B) and (B, A) , with action A corresponding to yielding and action B to driving straight. The Chicken moral game supports a rich set of subgame outcomes across the parameter space. High individualism supports the material game Nash equilibria. High collectivism with equality supports the homogeneous efficient outcome (A, A) , while high collectivism with inequality supports the heterogeneous efficient outcome favoring the more influential player. In intermediate parameter regions, both the homogeneous and heterogeneous efficient outcomes are simultaneously supported.

E Inverse Chicken Game

Table 17: Payoff matrix for the Inverse Chicken game

	A	B
A	-1, 1	0, 0
B	-10, -10	1, -1

Figure 15 shows the supported subgame outcomes across moral-weight parameters. The Inverse Chicken material game reverses Player 2's action labels relative to the Chicken game, so that Player 1's action A still corresponds to yielding and action B to driving straight, while for Player 2, action A now corresponds to driving straight and action B to yielding;

¹⁴The SPNE graph at $W = 8$ exhibits numerical instability in the region of high collectivism and low inequality, producing discontinuous patches that are likely artefacts of the algorithm. The true equilibrium correspondence in that region is suspected to be empty (coded 0).

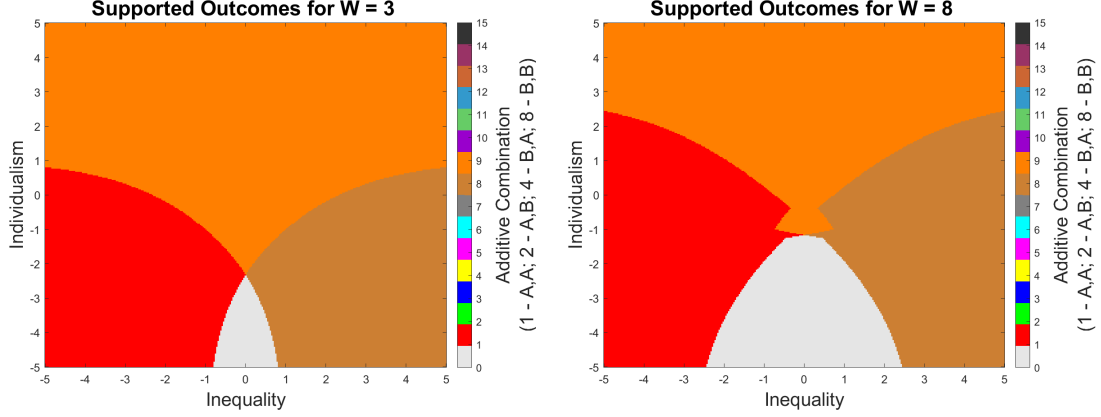


Figure 15: Supported subgame outcomes in the Inverse Chicken moral game ($W = 3, 8$ left to right)

the two Nash equilibria of the material game are the homogeneous outcomes (A, A) and (B, B) . The Inverse Chicken moral game supports equilibrium regions qualitatively similar to those of the Chicken moral game. High individualism supports the material game Nash equilibria, while high inequality uniquely implements the heterogeneous efficient outcome favoring the more influential player. Under high collectivism and equality, however, no pure-strategy equilibria exist. The egalitarian material game efficient outcome (A, B) , being heterogeneous under the inverse labeling, requires minimal differentiation in value judgments across actions for incentive compatibility, yet such minimal differentiation is inconsistent with supporting this outcome. The homogeneous efficient outcomes remain unsupported because the disadvantaged player retains sufficient projecting moral weight to induce the alternative efficient outcome.

F Zero-Sum Game

Table 18: Payoff matrix for the Zero-Sum game

	A	B
A	1, -1	-1, 1
B	-1, 1	1, -1

Figure 16 shows the supported subgame outcomes across moral-weight parameters. The Zero-Sum material game has no pure-strategy Nash equilibrium and is not symmetric. The SPNE existence requirement for off-path pure-strategy subgame equilibria eliminates the possibility of supporting any outcome for most moral-weight configurations. Homogeneous

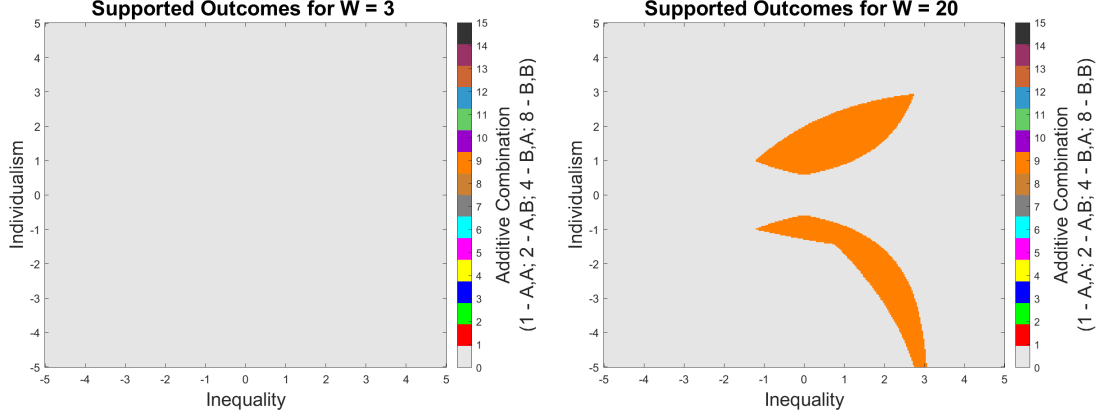


Figure 16: Supported subgame outcomes in the Zero-Sum moral game ($W = 3, 20$ left to right)

outcomes emerge as moral weight scales up.

B Proofs

Proof of Lemma 1. Assume there exists RTC SPNE

$$\{(\alpha_1^*, \beta_1^*, f_1(m_1, m_2)), (\alpha_2^*, \beta_2^*, f_2(m_1, m_2))\}$$

where both $\alpha_i^*, \beta_i^* < 1$ for $i \in \{1, 2\}$. Let $\delta = 1 - \max\{\alpha_i^*, \beta_i^*\}$. There exists a profitable deviation $\tilde{\alpha}_i = \alpha_i^* + \delta$, $\tilde{\beta}_i = \beta_i^* + \delta$ for player i , such that the original subgame outcome is still supported. By the RTC SPNE rule, the on-path subgame outcome remains the same, yet player i 's utility is strictly higher. This is a contradiction.

If subgame outcome (A, A) is supported by any morality, then $\alpha_1^* = \alpha_2^* = 1$ and $\beta_1^* = \beta_2^* = 0$ must support it as well, and provides the maximum amount of social payoff for both players while leaving the subgame outcome and thus the material payoffs unchanged. Thus, both players have an incentive to set $\alpha_1^* = \alpha_2^* = 1$. The same logic applies for subgame outcome (B, B) . $\square$

Proof of Theorem 1. By Lemma 1, for homogeneous actions to be in an RTC SPNE, both players must receive a value judgment of 1 for their on-path action. Since utility is the sum of material payoff and social payoff, and the pair of actions is Pareto efficient for the material game, it has to be Pareto efficient for the moral game. $\square$

Proof of Proposition 1. For any moral weights, all outcomes can be supported when both players set value judgments $\alpha_i = \beta_i = 1$ for all i . $\square$

Proof of Theorem 2. I construct an RTC SPNE supporting (A, A) with scaled weights $w'_{ij} = kw_{ij}$; the construction for (B, B) is symmetric.

On path, each player i sets $\alpha_i = 1$, $\beta_i = 0$ and plays A . Off path, players play A whenever (A, A) is a subgame Nash equilibrium given the prevailing moralities, and otherwise play any subgame Nash equilibrium.

With $\alpha_i = 1$ and $\beta_i = 0$, the subgame Nash conditions are satisfied when

$$k \geq k_{\text{sub}} \equiv \max \left\{ \frac{u_1^M(B, A) - u_1^M(A, A)}{w_{11} + w_{12}}, \frac{u_2^M(A, B) - u_2^M(A, A)}{w_{21} + w_{22}} \right\}.$$

I verify that player 1 has no profitable first-stage deviation; the argument for player 2 is symmetric. For deviations inducing (B, A) or (B, B) , since $\beta_2 = 0$, player 1 loses social payoff w'_{12} while gaining at most $\max\{u_1^M(B, A), u_1^M(B, B)\} - u_1^M(A, A)$ in material payoff. Such a deviation is unprofitable when

$$k \geq k_1 \equiv \frac{\max\{u_1^M(B, A) - u_1^M(A, A), u_1^M(B, B) - u_1^M(A, A)\}}{w_{12}}.$$

For a deviation inducing (A, B) , by Lemma 1 player 1 optimally sets $\beta'_1 = 1$ and chooses the highest α'_1 consistent with player 2 preferring B . The binding constraint $u_2^M(A, B) + w'_{21} \geq u_2^M(A, A) + w'_{21}\alpha'_1 + w'_{22}$ yields $1 - \alpha'_1 = (u_2^M(A, A) - u_2^M(A, B) + w'_{22})/w'_{21}$. Player 1's self-approval loss is $(1 - \alpha'_1)w'_{11}$, so this deviation is unprofitable when

$$k \geq k_2 \equiv \frac{u_1^M(A, B) - u_1^M(A, A) - \frac{w_{11}}{w_{21}}(u_2^M(A, A) - u_2^M(A, B))}{\frac{w_{11}w_{22}}{w_{21}}}.$$

Let $K_1^{AA} = \max\{k_{\text{sub}}, k_1, k_2\}$ for player 1's deviations when supporting (A, A) , and define K_2^{AA} symmetrically. Define K_1^{BB} and K_2^{BB} analogously for supporting (B, B) with $\beta_i = 1$, $\alpha_i = 0$. Set $k' = \max\{K_1^{AA}, K_2^{AA}, K_1^{BB}, K_2^{BB}\}$. For any $k \geq k'$, both (A, A) and (B, B) are supported as RTC SPNE. $\square$

Proof of Corollary 1. Follows from Theorem 2 and Theorem 1. $\square$

Proof of Proposition 2. For homogeneous actions under RTC SPNE, efficiency is given by Lemma 1 and Theorem 1. The indifference moral game must have $\alpha_1 = \beta_1 = \alpha_2 = \beta_2 = 1$ in order to support heterogeneous actions in an RTC SPNE. Both players must assign a value judgment of 1 to their own action to maximize self-righteous payoff. If either player assigns a value judgment strictly below 1 to the other player's action, the other can deviate to play the same action and assign a value judgment of 1, strictly improving utility and violating SPNE.

Since all value judgments are 1, and any heterogeneous actions are Pareto efficient in the indifference material game, by additivity of utility, any RTC SPNE is Pareto efficient. $\square$

Proof of Theorem 3. I prove the contrapositive. Suppose (A, B) is supported as a pure-strategy RTC SPNE. I derive a contradiction for each of the four cases in which a homogeneous outcome strictly Pareto dominates a heterogeneous one. The four cases are (A, A) strictly Pareto dominating (A, B) (Case 1), (A, A) strictly Pareto dominating (B, A) (Case 2), (B, B) strictly Pareto dominating (A, B) (Case 3), and (B, B) strictly Pareto dominating (B, A) (Case 4). Case 1 is treated in full; the remaining cases follow by symmetric arguments.

Case 1. Suppose (A, A) strictly Pareto dominates (A, B) .

Let $D_1 = u_1^M(A, A) - u_1^M(B, A) > 0$ and $D_2 = u_2^M(A, A) - u_2^M(A, B) > 0$. Suppose for contradiction that (A, B) is supported as an RTC SPNE with on-path moralities $(\alpha_1, \beta_1, \alpha_2, \beta_2)$ and $\delta_i = \alpha_i - \beta_i \in [-1, 1]$. The subgame Nash conditions are

$$u_1^M(A, B) - u_1^M(B, B) + w_{11}\delta_1 + w_{12}\delta_2 \geq 0, \quad (\text{B1})$$

$$u_2^M(A, B) - u_2^M(A, A) - w_{21}\delta_1 - w_{22}\delta_2 \geq 0. \quad (\text{B2})$$

By strict Pareto dominance, (B2) gives

$$w_{21}\delta_1 + w_{22}\delta_2 \leq -D_2 < 0. \quad (\text{B3})$$

By Lemma 1, $\max\{\alpha_1, \beta_1\} = 1$ and $\max\{\alpha_2, \beta_2\} = 1$. I consider all four sub-cases.

Sub-case (i): $\alpha_1 = 1$ and $\alpha_2 = 1$. Then $\delta_1 \geq 0$ and $\delta_2 \geq 0$, so $w_{21}\delta_1 + w_{22}\delta_2 \geq 0$, contradicting (B3).

Sub-case (ii): $\alpha_1 = 1$ and $\beta_2 = 1$. Then $\delta_1 \geq 0$ and $\delta_2 \leq 0$. Player 2 deviates to $\delta'_2 = 1$ (setting $\alpha'_2 = 1, \beta'_2 = 0$).

Under moralities $(\delta_1, 1)$, the outcome (A, B) is not Nash. Player 2's Nash condition for (A, B) would require $w_{21}\delta_1 + w_{22} \leq -D_2$, but $w_{21}\delta_1 + w_{22} \geq w_{22} > 0 > -D_2$. By the RTC rule, the subgame outcome switches away from (A, B) .

Moreover, (A, A) is the unique Nash under $(\delta_1, 1)$. Player 1 prefers A since $D_1 + w_{11}\delta_1 + w_{12} \geq D_1 + w_{12} > 0$ (using $\delta_1 \geq 0$). Player 2 prefers A since $D_2 + w_{21}\delta_1 + w_{22} \geq D_2 + w_{22} > 0$. Outcome (B, A) would require $w_{11}\delta_1 + w_{12} \leq -D_1$, but $w_{11}\delta_1 + w_{12} \geq w_{12} > 0 > -D_1$. Outcome (B, B) would require $w_{11}\delta_1 + w_{12} \leq u_1^M(B, B) - u_1^M(A, B)$, but from (B1) and $\delta_2 \leq 0 < 1$, $w_{11}\delta_1 + w_{12} > w_{11}\delta_1 + w_{12}\delta_2 \geq u_1^M(B, B) - u_1^M(A, B)$. Hence (A, A) is the unique Nash, and RTC selects it.

The deviation is profitable for player 2. On path, player 2 plays B at (A, B) , earning social payoff $w_{21}\beta_1 + w_{22}$ (since $\beta_2 = 1$). After the deviation, player 2 plays A at (A, A) , earning $w_{21} + w_{22}$ (since $\alpha_1 = 1$ and $\alpha'_2 = 1$). The total gain is

$$\underbrace{u_2^M(A, A) - u_2^M(A, B)}_{=D_2>0} + \underbrace{w_{21}(1 - \beta_1)}_{\geq 0} > 0,$$

where $\beta_1 \in [0, 1]$ since $\alpha_1 = 1$, and the self-weight w_{22} is unchanged because $\beta_2 = 1$ on path and $\alpha'_2 = 1$ after the deviation. This is a contradiction.

Sub-case (iii): $\beta_1 = 1$ and $\alpha_2 = 1$. Then $\delta_1 \leq 0$ and $\delta_2 \geq 0$. Player 1 deviates to $\delta'_1 = 1$ (setting $\alpha'_1 = 1, \beta'_1 = 0$).

Under moralities $(1, \delta_2)$, the outcome (A, B) is not Nash, since player 2's condition would require $w_{21} + w_{22}\delta_2 \leq -D_2$, but $w_{21} + w_{22}\delta_2 \geq w_{21} > 0 > -D_2$.

Moreover, (A, A) is the unique Nash under $(1, \delta_2)$. Player 1 prefers A since $D_1 + w_{11} + w_{12}\delta_2 \geq D_1 + w_{11} > 0$. Player 2 prefers A since $D_2 + w_{21} + w_{22}\delta_2 \geq D_2 + w_{21} > 0$. Outcome (B, A) would require $w_{11} + w_{12}\delta_2 \leq -D_1$, but $w_{11} + w_{12}\delta_2 \geq w_{11} > 0 > -D_1$. Outcome (B, B) would require $w_{11} + w_{12}\delta_2 \leq u_1^M(B, B) - u_1^M(A, B)$, but from (B1) and $\delta_1 \leq 0$, $w_{11} + w_{12}\delta_2 > w_{11}\delta_1 + w_{12}\delta_2 \geq u_1^M(B, B) - u_1^M(A, B)$.

The deviation is profitable for player 1. With $\alpha_1 = \delta_1 + 1 \in [0, 1]$ (since $\beta_1 = 1$), the gain is

$$\underbrace{u_1^M(A, A) - u_1^M(A, B)}_{>0} + \underbrace{w_{11}(1 - \alpha_1)}_{\geq 0} > 0,$$

where $u_1^M(A, A) > u_1^M(A, B)$ by strict Pareto dominance. This is a contradiction.

Sub-case (iv): $\beta_1 = 1$ and $\beta_2 = 1$. Then $\delta_1 \leq 0$ and $\delta_2 \leq 0$. I handle two sub-sub-cases.

Sub-sub-case (iv-a): $\delta_1 < 0$, so $\alpha_1 < 1$. Player 1 deviates to $\delta'_1 = 1$ (setting $\alpha'_1 = 1, \beta'_1 = 0$).

If (A, B) remains Nash under $(1, \delta_2)$, then by the RTC rule the subgame outcome stays at (A, B) , and player 1's payoff increases by $w_{11}(1 - \alpha_1) > 0$. This is a contradiction.

Now suppose (A, B) is not Nash under $(1, \delta_2)$. From (B1) with $\delta_1 < 0$ and $\delta_2 \leq 0$, I have $u_1^M(A, B) > u_1^M(B, B)$. I show that (B, B) is also not Nash under $(1, \delta_2)$. For player 1 to prefer B at (B, B) it would be necessary that

$$u_1^M(B, B) - u_1^M(A, B) \geq w_{11} + w_{12}\delta_2 = w_{11} - w_{12}|\delta_2|. \quad (\text{B4})$$

But from (B1), $u_1^M(B, B) - u_1^M(A, B) \leq -(w_{11}|\delta_1| + w_{12}|\delta_2|)$. Substituting into (B4) gives $-w_{11}|\delta_1| \geq w_{11}$, i.e., $|\delta_1| \leq -1$, which is impossible. Hence (B, B) is not Nash under $(1, \delta_2)$.

Since (A, B) and (B, B) are both ruled out, the Nash outcome under $(1, \delta_2)$ lies in $\{(A, A), (B, A)\}$. Because (A, B) is assumed to be a pure-strategy RTC SPNE, every off-path morality profile induces a subgame with a pure Nash equilibrium, so at least one of (A, A) and (B, A) is a Nash equilibrium under $(1, \delta_2)$. Let $G = [u_1^M(A, A) - u_1^M(A, B)] + w_{11}(1 - \alpha_1)$. Since (A, A) strictly Pareto dominates (A, B) and $\alpha_1 \in [0, 1)$, I have $G > 0$. I show that player 1 gains strictly under either possible selection.

If the RTC rule selects (A, A) , then on-path player 1 earns $u_1^M(A, B) + w_{11}\alpha_1 + w_{12}\alpha_2$, while after the deviation player 1 earns $u_1^M(A, A) + w_{11} + w_{12}\alpha_2$. The gain equals $G > 0$.

If the RTC rule selects (B, A) , then (B, A) being Nash for player 1 requires $-D_1 - w_{11} - w_{12}\delta_2 \geq 0$, which rearranges to $w_{12}(1 - \alpha_2) \geq D_1 + w_{11}$ (using $\delta_2 = \alpha_2 - 1$). On-path player 1 earns $u_1^M(A, B) + w_{11}\alpha_1 + w_{12}\alpha_2$, while after the deviation player 1 earns $u_1^M(B, A) + w_{12}$. The gain satisfies

$$\begin{aligned} \text{Gain} &= u_1^M(B, A) - u_1^M(A, B) + w_{12}(1 - \alpha_2) - w_{11}\alpha_1 \\ &\geq u_1^M(B, A) - u_1^M(A, B) + (D_1 + w_{11}) - w_{11}\alpha_1 \\ &= \underbrace{[u_1^M(B, A) + D_1]}_{=u_1^M(A, A)} - u_1^M(A, B) + w_{11}(1 - \alpha_1) \\ &= G > 0. \end{aligned}$$

In either case player 1's deviation is strictly profitable. This is a contradiction.

Sub-sub-case (iv-b): $\delta_1 = 0$ and $\delta_2 < 0$, so $\alpha_1 = \beta_1 = 1$ and $\alpha_2 < 1$. Player 1 has no morality gain available, so player 2 deviates to $\delta'_2 = 1$ (setting $\alpha'_2 = 1$, $\beta'_2 = 0$).

Under moralities $(0, 1)$, the outcome (A, B) is not Nash since player 2's condition would require $w_{22} \leq -D_2$, but $w_{22} > 0 > -D_2$. Moreover, (A, A) is the unique Nash. Player 1's condition yields $D_1 + w_{12} > 0$ and player 2's condition yields $D_2 + w_{22} > 0$. Outcome (B, A) fails player 1's condition since $-D_1 - w_{12} < 0$. Outcome (B, B) fails player 1's condition since from (B1) with $\delta_1 = 0$ and $\delta_2 < 0$, $u_1^M(A, B) - u_1^M(B, B) \geq w_{12}|\delta_2| > 0$, so $u_1^M(B, B) - u_1^M(A, B) < 0 < w_{12}$.

The deviation is profitable for player 2. The material gain is $D_2 > 0$. The cross-weight gain is $w_{21}(\alpha_1 - \beta_1) = 0$ (since $\delta_1 = 0$), and the self-weight w_{22} is unchanged because $\beta_2 = 1$ on path and $\alpha'_2 = 1$ after the deviation. This is a contradiction.

In all sub-cases of Case 1, I obtain a contradiction.

Cases 2, 3, and 4.

The arguments are symmetric. In each case, Lemma 1 forces $\max\{\alpha_i, \beta_i\} = 1$ for each player i . The sub-case in which both players' extremal judgments align with their on-

path action (analogous to sub-case (i)) is ruled out by the relevant Nash condition. In the remaining sub-cases, one identifies a player whose extremal judgment is opposite to the on-path action (analogous to sub-cases (ii) and (iii)), or applies the on-path Nash conditions to rule out all outcomes except the Pareto-dominant one after the deviation (analogous to sub-case (iv)). In every case a profitable deviation exists without invoking any weight conditions.

In all four cases I reach a contradiction, so no heterogeneous outcome can be supported as a pure-strategy RTC SPNE when any homogeneous outcome strictly Pareto dominates it. $\square$

C Computational Algorithm for Moral Equilibrium Selection

A Overview

This appendix details the computational algorithm employed to identify subgame perfect Nash equilibria in the two-stage moral game. The algorithm systematically determines which of the four possible outcomes can be supported as SPNE. Rather than working directly with $\alpha_1, \alpha_2, \beta_1, \beta_2$, the algorithm parameterizes each player's morality by the difference $\delta_1 = \alpha_1 - \beta_1$ and $\delta_2 = \alpha_2 - \beta_2$. Lemma 1 provides the mapping from δ_i to the corresponding α_i and β_i values.

For a subgame outcome to be supported as an SPNE, two conditions must hold. The second-stage Nash equilibrium conditions must be satisfied, and no player can profitably deviate in the first stage.

For outcome (A, A) , the second-stage Nash equilibrium conditions require values of $\delta_1, \delta_2 \in [-1, 1]$ satisfying

$$\begin{aligned} u_1^M(A, A) - u_1^M(B, A) + w_{11}\delta_1 + w_{12}\delta_2 &\geq 0 \\ u_2^M(A, A) - u_2^M(A, B) + w_{21}\delta_1 + w_{22}\delta_2 &\geq 0 \end{aligned}$$

For outcome (A, B) , the conditions are

$$\begin{aligned} u_1^M(A, B) - u_1^M(B, B) + w_{11}\delta_1 + w_{12}\delta_2 &\geq 0 \\ u_2^M(A, B) - u_2^M(A, A) - w_{21}\delta_1 - w_{22}\delta_2 &\geq 0 \end{aligned}$$

For outcome (B, A) , the conditions are

$$\begin{aligned} u_1^M(B, A) - u_1^M(A, A) - w_{11}\delta_1 - w_{12}\delta_2 &\geq 0 \\ u_2^M(B, A) - u_2^M(B, B) + w_{21}\delta_1 + w_{22}\delta_2 &\geq 0 \end{aligned}$$

For outcome (B, B) , the conditions are

$$\begin{aligned} u_1^M(B, B) - u_1^M(A, B) - w_{11}\delta_1 - w_{12}\delta_2 &\geq 0 \\ u_2^M(B, B) - u_2^M(B, A) - w_{21}\delta_1 - w_{22}\delta_2 &\geq 0 \end{aligned}$$

When such values of δ_1 and δ_2 exist for a given outcome, the algorithm proceeds to test for profitable deviations.

B Feasible Region Construction

The algorithm constructs the feasible region differently depending on whether the outcome is homogeneous or heterogeneous.

Homogeneous Outcomes (A, A) and (B, B) . For homogeneous outcome (A, A) , the algorithm tests the single point where $\delta_1 = 1$ and $\delta_2 = 1$, which corresponds to both players assigning maximal value judgment to action A . Similarly, for outcome (B, B) , the algorithm tests the point $\delta_1 = -1$ and $\delta_2 = -1$, corresponding to both players maximally praising action B .

Heterogeneous Outcomes (A, B) and (B, A) . For heterogeneous outcomes, the algorithm searches the full domain $\delta_1 \in [-1, 1]$ and $\delta_2 \in [-1, 1]$ to identify all moral configurations satisfying the subgame Nash equilibrium constraints.

Within the full domain, the two linear inequality constraints define a convex polygonal feasible region. To systematically identify test points, the algorithm partitions the domain into four quadrants and proceeds through five stages. *Candidate generation* produces corner points comprising the four domain corners $(\pm 1, \pm 1)$, intersections of the two constraint lines with the domain boundaries ($\delta_1 = \pm 1$ and $\delta_2 = \pm 1$) and with the interior axes ($\delta_1 = 0$ and $\delta_2 = 0$), the origin $(0, 0)$, and the intersection of the two constraint lines with each other. *Feasibility filtering* retains only candidates that satisfy both inequality constraints and lie within $[-1, 1] \times [-1, 1]$. *Quadrant partitioning* assigns feasible candidates to four quadrants, Q1 ($\delta_1 \in [-1, 0]$, $\delta_2 \in [-1, 0]$), Q2 ($\delta_1 \in [-1, 0]$, $\delta_2 \in [0, 1]$), Q3 ($\delta_1 \in [0, 1]$, $\delta_2 \in [-1, 0]$), and Q4 ($\delta_1 \in [0, 1]$, $\delta_2 \in [0, 1]$), where boundary points may belong to multiple quadrants. *Convex hull construction* then, for each quadrant with at least three feasible points, builds the convex hull and extracts its vertices; quadrants with fewer than three points retain all

feasible points; vertices from all quadrants are combined and deduplicated to form the final test set. *Perturbation enhancement* augments each hull vertex with small perturbations of magnitude $\epsilon = 10^{-7}$ toward neighboring vertices along hull edges, adds perpendicular perturbations for vertices on an axis ($\delta_1 = 0$ or $\delta_2 = 0$), and inserts axis intersection points plus perturbations for hull edges crossing an axis. For outcome (B, A) , the algorithm applies the same full-domain procedure with appropriately adjusted constraint inequalities.

C Deviation Testing

Once the test points are identified, the algorithm verifies that no player can profitably deviate. For homogeneous outcomes a single extreme point is tested; for heterogeneous outcomes the algorithm works through all hull vertices with their perturbations. Each test point (δ_1, δ_2) is assessed in two stages.

Subgame verification computes utilities for all four outcomes under the candidate moral configuration. If the candidate point does not yield the target outcome as a Nash equilibrium, it is immediately rejected. Nash equilibrium conditions use standard inequality checks with a geometric tolerance of 10^{-10} .

Deviation testing then examines whether either player can profitably deviate. For Player 1, the algorithm fixes δ_2 and constructs the set of candidate deviations δ_1 , comprising the boundary values $\delta_1 \in \{-1, 1\}$, values derived from the intervals where each outcome becomes a Nash equilibrium given δ_2 , and small perturbations of $\pm 10^{-8}$ around these critical values. For each candidate deviation, the algorithm computes Player 1's utility across all subgame Nash equilibria under the new moral configuration. If no subgame Nash equilibrium exists at the deviated moral profile, the deviation is immediately flagged as profitable, since SPNE requires every off-path moral profile to induce a well-defined subgame outcome. Otherwise, the algorithm operates in one of two modes. In standard SPNE mode, Player 1's payoff from the deviation is the minimum across all valid subgame outcomes, reflecting worst-case equilibrium selection. In RTC SPNE mode, if the on-path outcome remains a Nash equilibrium under the deviated profile, it is selected as the subgame outcome; otherwise the minimum applies. A deviation is profitable if the resulting payoff exceeds Player 1's current payoff by more than a tolerance of 2×10^{-12} . Player 2's deviation test is symmetric, fixing δ_1 and searching over alternative values of δ_2 .

When any test point passes the subgame verification and finds no profitable deviation for either player, the outcome is confirmed as supportable in SPNE. For homogeneous outcomes, only one point is tested. For heterogeneous outcomes, the algorithm terminates the search as soon as one such point is found.

D Computational Efficiency

The algorithm achieves computational efficiency by focusing on vertices of the feasible region rather than exhaustively searching the interior. This approach exploits the linearity of both the Nash equilibrium constraints and the utility functions. Any SPNE must occur at extreme points of the feasible region, as interior points cannot be optimal given the linear structure. By testing only the corner points, the algorithm comprehensively checks all potentially optimal moral configurations while minimizing computational burden.

E Parameter Sweep Implementation

The main computational framework loops over three parameters, total moral weight W , inequality parameter Ine , and individualism parameter Ind . For each parameter combination, the algorithm (i) converts (W, Ine, Ind) to individual weights $(w_{11}, w_{21}, w_{12}, w_{22})$ using the exponential transformation specified in the main text, (ii) calls the outcome testing procedure for each of the four outcomes, (iii) encodes the results as a sum where each of the four outcomes contributes a specific power of two if it is supportable, specifically $1 \cdot (A, A)_{true} + 2 \cdot (A, B)_{true} + 4 \cdot (B, A)_{true} + 8 \cdot (B, B)_{true}$ where each indicator is 1 if the outcome is supportable and 0 otherwise, and (iv) stores this encoded value in a three-dimensional output matrix.

This encoding scheme allows each integer from 0 to 15 to uniquely represent which subset of the four outcomes can be supported as SPNE for given parameter values. A value of 9 indicates that outcomes (A, A) and (B, B) are supportable, while a value of 15 indicates all four outcomes are supportable.

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